

1. Consider a copper wire of cross-section area 1 mm^2 carrying a current 1A . What is the drift velocity of the electron? The density and molecular weight of Cu are 9 gm/cm^3 and 64 gm/mole respectively.
Ans: $7 \times 10^{-5} \text{ m/sec}$
2. Copper has FCC structure and the atomic radius is $1.278 \text{ \AA}$. Calculate density of copper crystal.
Ans: 8940 Kg/m^3
3. NaCl crystal have FCC structure. The density of sodium Chloride is 2180 Kg/ m^3 . Calculate the distance between two adjacent atoms. Atomic weight of Sodium = 23 a.m.u and that of Chloride = 35.5 a.m.u .
Ans 0.2828 nm
4. Calculate the energy required for the formation of NaCl ionic molecule if energy required for formation of Na^+ and Cl^- ions is 5.14 eV and 3.62 eV respectively. For $2.51 \text{ \AA}$ be the equilibrium separation constant between Na^+ and Cl^- ?
Ans: 4.22 eV
5. The dissociation energy of KF molecule is 15.12 eV . The ionization energy of K is 4.34 eV , and the electron affinity of F is 4.07 eV . What is the equilibrium separation constant for the KF molecule?
Ans: $2.675 \text{ \AA}$
6. A substance with face centered cubic lattice has density 6250 Kg/ m^3 and molecular weight 60.2 . Calculate the lattice constant 'a' of crystal. Given Avogadro number = $6.02 \times 10^{26} \text{ (kg mole)}^{-1}$
Ans: 0.4 nm
7. The energy of two particles in the field of each other at a separation r is given by $U = \frac{\alpha}{r} + \frac{\beta}{r^8}$;
a and b are constants. At what separation they will form a stable compound?
Ans $r = \left(\frac{-8\beta}{\alpha}\right)^{\frac{1}{7}}$
8. The energy gap of some alkali halides are KCl = 7.6 eV , KBr = 6.3 eV , KI = 5.6 eV . Which of these are transparent to visible light? At what wave length does each become opaque?
Ans: ($380\text{-}750$) nm
9. Consider a current carrying copper wire, the number of free electrons in copper is $8.4 \times 10^{28} \text{ electron /m}^3$. a. Calculate the Fermi energy for Cu.
Ans: 6.95 eV
b. At what temperature, T_f will the average thermal energy $K_B T_f$ of a gas be equal to that energy
Ans: $80,500 \text{ K}$
10. The density of aluminium is 2.70 gm cm^{-3} and its molecular weight is $26.98 \text{ gm mole}^{-1}$. a. Calculate the Fermi energy.
Ans: 11.66 eV
b. If the experimental values of E_f is 12 eV , what is the electron effective mass in aluminum? Aluminium is trivalent.
Ans: $0.97 m_e$
11. There are 2.5×10^{28} free electrons per cubic meter of sodium. Calculate the Fermi energy and Fermi velocity.
Ans: 3.1 eV and $1.04 \times 10^6 \text{ ms}^{-1}$
12. Calculate silver is the metallic state with one free electron per atom. Calculate the Fermi energy, given density of silver is 10.5 gm cm^{-3} and atomic weight 108 .
Ans 5.48 eV
13. Assuming that, each atom of copper contributes one free electron calculate the drift velocity of free electrons in copper conductor of cross sectional area $2 \times 10^{-4} \text{ m}^2$ carrying current of 100A . [Atomic wt. Of copper = 63.5 u and density of copper = $8.94 \times 10^3 \text{ Kg m}^{-3}$]
Ans: $3.68 \times 10^{-5} \text{ ms}^{-1}$.

NOTE: Dissociation energy / Binding energy = Electron affinity - ionization energy – potential energy.

$$\text{Potential energy} = - \frac{e^2}{4 \pi \epsilon_0 r}$$